

OPERATING SYSTEM AND SYSTEM PROGRAMMING(OSSP)

NAME: K. SHASHANK REDDY

SECTION: 08

ROLL NO: 2520030215

SKILL - 1

1. Aim: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing

```

mrshashank@LAPTOP-203MS  ×  +  ▾
make --version
git --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
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warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

GNU Make 4.4.1
Built for x86_64-pc-linux-gnu
Copyright (C) 1988-2023 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.htm
l>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
git version 2.53.0
mrshashank@LAPTOP-203MS5J7:~$ mkdir ossp-shell
cd ossp-shell
mrshashank@LAPTOP-203MS5J7:~/ossp-shell$ mkdir src include bin
mrshashank@LAPTOP-203MS5J7:~/ossp-shell$ tree
Command 'tree' not found, but can be installed with:
sudo snap install tree # version 2.1.3+pkg-5852, or
sudo apt install tree # version 2.3.1-1
See 'snap info tree' for additional versions.
mrshashank@LAPTOP-203MS5J7:~/ossp-shell$ sudo apt install tree
[sudo: authenticate] Password:
Installing:
  tree

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 54
  Download size: 53.5 kB
  Space needed: 125 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 tree amd64 2.
3.1-1 [53.5 kB]
Fetched 53.5 kB in 1s (41.9 kB/s)
Selecting previously unselected package tree.
(Reading database ... 42276 files and directories currently installed.)
Preparing to unpack .../tree_2.3.1-1_amd64.deb ...
Unpacking tree (2.3.1-1) ...
Setting up tree (2.3.1-1) ...

```

```

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 54
  Download size: 53.5 kB
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Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 tree amd64 2
3.1-1 [53.5 kB]
Fetched 53.5 kB in 1s (41.9 kB/s)
Selecting previously unselected package tree.
(Reading database ... 42276 files and directories currently installed.)
Preparing to unpack .../tree_2.3.1-1_amd64.deb ...
Unpacking tree (2.3.1-1) ...
Setting up tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
mrshashank@LAPTOP-203MS5J7:~/ossp-shell$ nano src/main.c
mrshashank@LAPTOP-203MS5J7:~/ossp-shell$ gcc src/main.c -o bin/shell
mrshashank@LAPTOP-203MS5J7:~/ossp-shell$ ./bin/shell
OSSP Shell Started
mrshashank@LAPTOP-203MS5J7:~/ossp-shell$ nano Makefile

```

```
execve("/usr/bin/lis", ["lis"], 0x7ff095dfe6e0 /* 26 vars */) = 0  
brk(NULL) = 0x62d2e39ea000  
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS,  
-1, 0) = 0x779fa2423000  
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)  
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=20887, ...}) = 0  
mmap(NULL, 20887, PROT_READ, MAP_PRIVATE, 3, 0) = 0x779fa241e000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3  
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0...  
832") = 832  
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0  
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x779fa23eb000  
mmap(0x779fa23f2000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x779fa23f2000  
mmap(0x779fa2413000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x779fa2413000  
mmap(0x779fa241a000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x779fa241a000  
mmap(0x779fa241c000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x779fa241c000  
close(3) = 0  
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3  
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0...  
832") = 832  
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0  
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x779fa23bd000  
mmap(0x779fa23c1000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x779fa23c1000  
mmap(0x779fa23e5000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x779fa23e5000  
mmap(0x779fa23e9000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MA
```

```

|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x62d2e3a4d000) = 0x62d2e3a4d000
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0
|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
ioctl(1, TIOCGWINSZ, {ws_row=40, ws_col=63, ws_xpixel=0, ws_ypi
xel=0}) = 0
ioctl(1, TIOCGWINSZ, {ws_row=40, ws_col=63, ws_xpixel=0, ws_ypi
xel=0}) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0
|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0
|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|IC
ANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
statx(AT_FDCWD, ".", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mas
k=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|07
55, stx_size=4096, ...}) = 0
statx(AT_FDCWD, ".", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mas
k=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|07
55, stx_size=4096, ...}) = 0
openat(AT_FDCWD, ".", O_RDONLY|O_NONBLOCK|O_CLOEXEC|O_DIRECTORY
) = 3
fstat(3, {st_mode=S_IFDIR|0755, st_size=4096, ...}) = 0
newfstatat(AT_FDCWD, ".", {st_mode=S_IFDIR|0755, st_size=4096,
...}, 0) = 0
getdents64(3, 0x62d2e3a4e150 /* 13 entries */, 32768) = 384
getdents64(3, 0x62d2e3a4e150 /* 0 entries */, 32768) = 0
close(3) = 0
write(1, "Makefile build exec_demo inc...", 100Makefile b
uild exec_demo include shell tests
README.md docs fork_demo parent_child src
) = 100
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NU
LL) = 0
munmap(0x779fa2420000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++

```

CODE:

```

#include <stdio.h>
#include <unistd.h>

int main()
{
    int pid = fork();

    if(pid == 0)
        printf("I am Child Process\n");
    else
        printf("I am Parent Process\n");

    return 0;
}

```

```
#include <stdio.h>

int main()
{
    printf("Linux Shell Project Started!\n");
    return 0;
}
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    int pid = fork();

    if(pid == 0)
        printf("I am Child Process\n");
    else
        printf("I am Parent Process\n");

    return 0;
}
```

```
#include <unistd.h>

int main()
{
    execl("/bin/ls", "ls", NULL);
    return 0;
}
```